Quantile Regression

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Introduction

This week we've learned about a semi-parametric alternative to linear regression that does not assume normally distributed errors. We've also seen how the quantile approach can be extended to ask more nuanced questions about the data: moving beyond *"Does X have a relationship with Y?"* to *"How does the relationship between X and Y vary?"*.

In this session, we'll put this into practice using an educational dataset on reading abilities. It is well-established that successful reading depends on both reading accuracy and language comprehension. While reading accuracy is a major constraint for beginning readers, we expect it to account for less variance in more skilled readers once those skills are fluent. In turn, variability in language skills might come to account for more of the variability in reading success.

We will use data from Year 8 pupils (age 12-13 years) to look at how these two variables—reading accuracy and language (vocabulary)—predict reading comprehension across the spectrum of reading ability. These data were collected as part of a larger project on reading comprehension (James et al., 2021).

Learning objectives

As a reminder, this week's learning objectives are to:

1. Explain the key assumptions and limitations of linear regression
2. Explain when and why quantile regression might be preferred
3. Describe and interpret quantile regression as a method for modelling the conditional median of an outcome
4. Explain how quantile regression uses weighted errors to model other parts of the outcome distribution
5. Compare and interpret regression coefficients across quantiles to understand how associations vary across the outcome distribution

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Setting up

Create a script

Start with the usual set-up steps for the practical sessions:

1. Open your ARM_Practicals.Rproj file in RStudio
2. Create a new script and save it in your scripts/ folder as **ARM4_quantile.R**
3. Download the data for the Week 4 practical from the VLE, and save it in your data/ folder.

> Tip
>
> If you don't have this folder on your current machine, repeat the set-up steps outlined in the Set-up steps from Week 1.

Load packages

We will need the *quantreg* package today's session, which you will first need to install using either the packages tab or by running `install.packages("quantreg")` in the Console.

Once you have done this, load the package at the top of your new script:

```
library(quantreg)
```

Load data

Now read in the data from your data/ folder.

```
read_dat <- read.csv("./data/ARM4_data_reading.csv")
```

The dataset consists of the following variables:

- **ppt_id:** a unique identifier for each participant
- **read_comp:** score on the reading comprehension test
- **read_acc:** composite score from the reading accuracy tests (Z-score)
- **vocab:** score on the vocabulary test (Z-score)

Our reading accuracy variable was Z-scored so that we could calculate a mean score across different tests (whereas averaging the original test scores would not have worked when the tests were on different scales). Z-scores produce a standardised score with a mean of 0 and standard deviation of 1. We did the same to the vocabulary measure so that our predictors will be on the same scale.

Remember that you can inspect the data by clicking the grid icon in the Environment window. Try calculating the mean and standard deviation of the different variables, as we did in Week 1.

```r
# Reading comprehension (outcome measure)
mean(read_dat$read_comp)
sd(read_dat$read_comp)

# Reading accuracy (Z-scored predictor)
mean(read_dat$read_acc)
sd(read_dat$read_acc)

# Vocabulary (Z-scored predictor)
mean(read_dat$vocab)
sd(read_dat$vocab)
```

Fit a linear model

Let's start by reminding ourselves how to fit a standard multiple regression model in R. We want to predict reading comprehension scores by reading accuracy and vocabulary as follows:

```
read_lm <- lm(read_comp ~ read_acc + vocab, data = read_dat)
```

Use the `summary()` function on your fitted model object to inspect the output. The `(Intercept)` term reflects the level of the outcome variable when all predictors are equal to 0. Our predictors were standardised so that their mean score was equal to 0 (as you verified above), which means we can interpret the intercept as the *average* comprehension score in the dataset.

Each coefficient estimate represents the change in comprehension score associated with a 1-unit change in the predictor. Again, we standardised these predictors so that 1 reflects a score 1 standard deviation above/below the sample's mean score. So we can interpret that being one standard deviation higher in the predictor is associated with the change in comprehension score indicated by the coefficient estimate.

Which predictor significantly predicted reading comprehension according to this model?

We should also check some relevant assumptions for this model. We can check the normality of the residuals by plotting a histogram as follows:

```
# Extract residuals
lm_resid <- resid(read_lm)

# Inspect distribution of residuals using histogram
hist(lm_resid)
```

The plot shows a mild negative skew (although would probably pass as acceptable).

> Tips for efficient code
>
> Note that we could do this more efficiently by directly nesting the arguments inside each other rather than creating new objects: (`hist(resid(read_lm))`. The trick here is to read/write the code from the inside out, so we first extract the residuals from the model, then put this directly inside our histogram function.

Fit a median regression model

A better alternative might be to fit a median regression model. We now use the `rq()` function to fit a quantile regression model instead, adding `tau = 0.5` to specify the 50th percentile. The formula otherwise stays the same.

```
# Fit median regression model
read_med <- rq(read_comp ~ read_acc + vocab, tau = 0.5, data = read_dat)

# Inspect output
summary(read_med)
```

How do the coefficients compare to your regular linear regression model? Note that the quantile regression approach doesn't provide precise *p*-values, but we can infer statistically significant relationships by inspecting with the 95% confidence intervals provided span 0[1].

We can alternatively ask for p-values when calling the `summary()` function, but need to specify how they should be calculated. One approach can be to include `se = "boot"` —we'll learn more about what that actually means next week!

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[1] If you would like the confidence intervals for your original model, you can get these by running `confint(read_lm)` .

Model multiple quantiles

Theories of reading predict that reading accuracy stops accounting for variance in reading comprehension in skilled readers, but that it might remain a limiting factor in struggling readers. We can test this possibility directly by looking at how our predictive relationships vary at different parts of the outcome distribution.

Let's look at the 10th and 90th percentiles, alongside our median. We can fit these all at once by using the concatenate function (`c()`) to list multiple values of tau:

```
# Fit model at three quantiles
read_rq <- rq(read_comp ~ read_acc + vocab, tau = c(0.1, 0.5, 0.9), data = read_dat)
```

Now use the `summary()` function to inspect the output as you did before. It should list three sets of model results, one for each quantile.

How do the predictors of reading comprehension change at the tail ends of the distribution?

> ⚠ Why do I get a warning message?
>
> R might print a warning that the solution may be nonunique. You can look this up online if you want to read more about it, but the general gist is that it isn't a problem of concern! You could explore this further by seeing which quantile the warning relates to (i.e., try fitting one quantile at a time, and see where the warning does/does not emerge.

Compare across quantiles

Finally, let's explore whether there are statistical differences between the quantiles. We can compare whether the slopes are different across quantiles using the `anova()` function. This asks whether the predictors have a similar relationship with the outcome variable across quantiles:

```
# Compare
anova(read_rq)
```

The results tell us that there is a significantly different relationship between the predictors and outcome variable between the different quantiles we fitted ($p = .01$). However, it doesn't tell us *where* those differences lie.

The plot function can be useful to inspect how each individual coefficients changes across quantiles, as different predictors might not change in the same way across the distribution. We can use the plot function to visualise this:

```
# Plot coefficients
plot(summary(read_rq))
```

The bottom right window should now show a plot for each component of the model. The x-axis for each plot is the quantile, and the y-axis is the value of the coefficient. The red lines mark the coefficient and confidence intervals for the regular (ordinary least squares) regression model. The black lines and shaded areas mark the coefficient and confidence intervals for the quantile regression model—linking between the three quantiles that we fitted in the model.

The intercept (i.e., average performance) increases steadily across quantiles, which makes sense considering we have weighted responses at different levels of reading comprehension. How do the two predictors of interest change across the distribution? This visualises the same points you reflected on above.

What have we learned about struggling readers that we would have missed by only carrying out the main regression analysis?

Summary

In this practical, we've explored how we might use quantile regression methods to (a) model the conditional median in cases where ordinary regression assumptions are not met; and (b) model differences in predictive relationships across different parts of the outcome distribution. From our ordinary regression, we might have concluded that there was no role for reading accuracy in reading comprehension, where the quantile regression highlighted this might be a relevant factor for struggling readers. Comparing regression coefficients across quantiles can give us richer insights into these relationships.

Challenge yourself

I have written some code to plot the relationship between reading accuracy and reading comprehension on a scatterplot, using a popular data visualisation package *ggplot2*. I have extracted and plotted the ordinary least squares regression line from our first model (`read_lm`) and then the median regression line from our quantile model (`read_rq`, tau = 0.5).

Start by copying and running this code.

```
# Load package for data visualisation
library(ggplot2)

# Specify dataframe and x/y variables
ggplot(read_dat, aes(x = read_acc, y = read_comp)) +

  # Add individual data points from dataset
  geom_point(shape = 1) +

  # Use theme for transparent background
  theme_classic() +

  # Add regression line from standard linear model
  geom_abline(intercept = coef(read_lm)[["(Intercept)"]],
              slope = coef(read_lm)[["read_acc"]],
              linetype = "longdash", size = 1) +

  # Add regression line from quantile model, tau = 0.5 (median)
  geom_abline(intercept = coef(read_rq)[["(Intercept)", "tau= 0.5"]],
              slope = coef(read_rq)[["read_acc", "tau= 0.5"]],
              colour = "blue", size = 1)
```

Now copy the section for plotting the median regression line twice more, and edit it to add the regression lines for tau = 0.1 and 0.9. You will need to add a `+` sign between each section so that the code continues to add each element to the plot. Try editing the colour or linetype to distinguish between your lines.

Once you're done, you could also copy and edit the code to plot the relationship between vocabulary and reading comprehension. Alternatively, you could apply everything we've done today to a new sample of Year 2 pupils, to see how these relationships look earlier in reading development (additional dataset on the VLE).

If you're interested in learning more about data visualisation in R, you might want to spend some time figuring out what each element contributes to the plot (there's also a whole DataCamp course you can try!).

Further learning

There is not a textbook chapter for this topic, but there are a few suggested background papers on the reading list that you might find helpful to consolidate your understanding (just don't get too hung up on elements that we haven't talked about!). A good alternative might be to find an example or two of quantile regression in a context that interests you.

> ⚠ How's it going?
>
> Let us know how today went via this quick survey - rate the difficulty and length, and leave any comments that would help for us to know (positive comments welcome too!). You will need to be logged in with your university account to respond, but your submission will be anonymous. For more general questions about the practical, please post them on the VLE Discussion Board.